Radio Standards

WS 2026

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1. Introduction

Organization

Lecture
Takes place on Mondays, 03:00 pm - 04:30 pm, in OEC3008 (except winter break)

Tutorial

- Takes place on Tuesdays, 11:00 am - 12:30 pm, in OEC3008 (except winter break)
- No tutorial in the classical sense, instead three group presentations about selected standards
- Learn to understand standards and improve presentation skills
- Topic and group selection via moodle (deadline next lecture)
- Digital video broadcasting (DVB), 0G-5G, WiFi, and Bluetooth not selectable, will be covered in this lecture
- Feel free to suggest interesting standards by yourself

Prerequisites

- Signal and system theory
- Basic engineering math

Lecture Material

- Please enroll in the Moodle course
- Lecture script will grow throughout the semester

Exam Regulations

Exam

- **30 min** oral exam
- Standards presented by your groups will be part of the exam

Bonus Points

Presentations give bonus points for the exam

Maximum: One-third grade step

e.g. $2.7 \rightarrow 2.3$, $1.3 \rightarrow 1.0$

Overall Goal

How to work with standards

Understand why standards exist	How to read them	How to understand them (on technical level)	How to make use of them

Electronic Measurements and Signal Processing Group EMS

We investigate the behavior of radio propagation channels and develop methods to make them usable.







- Development of measurement equipment for characterizing radio channels from 0 Hz to 300 GHz
- Measurement, simulation, and modeling of radio channels
- Analysis of radio channel characteristics using signal processing - high-resolution estimation methods and AI

- Exploitation of radio channel characteristics for environmental sensing - radar technology and integrated communications and sensing (ICAS)
- Emulation of electromagnetic wave propagation for testing mobile and satellite communications
- Standardization and development of solutions relevant to mobile communications standards

Radio Standards several decades ago...

... by Dr.-Ing. Alexander Ihlow

Topics to be Covered I

1. Technical Preliminaries
   - LTI Channel
   - Fourier Transformation
   - Spatial Processing
2. Standardization 101
   - How is the electromagnetic spectrum occupied?
   - What is a standard?
   - What is a standardization body?
   - What is a patent?
3. Propagation in a Nutshell (by Alexander Ihlow)
   - How do cable, satellite, and air interfaces differ from each other?
4. Evolution of Cellular Standards
   - 0G-2G
   - 3G
   - Deep dive OFDM
   - 4G
   - 5G

Topics to be Covered II

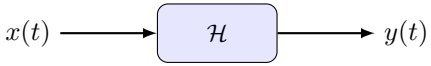
5. WiFi vs. Bluetooth vs. Cellular
   - Why do we have different standards for different applications and use cases?
6. Hands-on DVB (by Alexander Ihlow)
   - How to read a standard?
7. Business Models
   - How to commercialize a standard?
8. The Future of Mobile Communications
   - 6G and beyond
   - Trends and goals
   - Recent research at EMS

2. Preliminaries

Technical Preliminaries

LTI System Model

Channel as Block Model



- System operator $\mathcal{H}$
- Transforms input signal $x(t)$ to output signal $y(t)$

System Operator under linear time-invariant (LTI) Properties

- **L**inear
  - $\mathcal{H}\{\alpha x_1(t) + \beta x_2(t)\} = \alpha\mathcal{H}\{x_1(t)\} + \beta\mathcal{H}\{x_2(t)\}$
  - Superposition principle
- **T**ime **I**nvariant
  - $\mathcal{H}\{x(t-\tau)\} = y(t-\tau)$
  - Time delay τ of input equates to time delay τ of output

Remark: Real channels are inherently dynamic—covered in lecture 2 (coherence time)

Channel channel impulse response (CIR)

Response to an ideal Dirac pulse $\delta(t)$

$$h(t) = \mathcal{H}\{\delta(t)\}$$

LTI Input-Output Relation

Convolution of input $x(t)$ with CIR $h(t)$

$$y(t) = x(t) * h(t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,\mathrm{d}\tau$$

Fourier Transformation I

From Impulse Response to Transfer Function

- Time-domain CIR $h(t)$ describes channel delay behavior
- Fourier transform (FT) maps CIR to frequency domain channel transfer function (CTF)
- $h(t) \circ\!\!-\!\!\bullet\, H(f)$

Basis Functions & Eigenfunction Property

- Complex exponentials $\phi_f(t) = \mathrm{e}^{\mathrm{j}2\pi ft}$ serve as orthogonal basis (kernel)
- Exponentials pass through unchanged except for complex scaling
- $\mathcal{H}\{\mathrm{e}^{\mathrm{j}2\pi ft}\} = H(f) \cdot \mathrm{e}^{\mathrm{j}2\pi ft}$
- Complex scalar $H(f)$ gives channel attenuation and phase rotation at frequency f

FT (Analysis)

Decomposition into spectrum

$$H(f) = \int_{-\infty}^{\infty} h(t)\, \mathrm{e}^{-\mathrm{j}2\pi ft}\, \mathrm{d}t$$

inverse Fourier transform (IFT) (Synthesis)

Reconstruction of time signal

$$h(t) = \int_{-\infty}^{\infty} H(f)\, \mathrm{e}^{\mathrm{j}2\pi ft}\, \mathrm{d}f$$

Orthonormality of Complex Exponentials

$$\left\langle \mathrm{e}^{\mathrm{j}2\pi f_1 t}, \mathrm{e}^{\mathrm{j}2\pi f_2 t} \right\rangle = \delta(f_1 - f_2)$$

Fourier Transformation II

Time-Frequency Uncertainty

- Direct consequence of the FT scaling property
- $x(at) \circ\!\!-\!\!\bullet\ \frac{1}{|a|} X\left(\frac{f}{a}\right)$
- Narrow time duration $\implies$ wide spectral occupancy
- Time-limited signals can never be strictly band-limited

Impact on Radio Standards

- Sharp transitions (e.g., rect pulse) cause slow $1/f$ spectral decay
- High sidelobes leak energy thereby causing interference
- Regulatory spectral masks strictly limit out-of-band emissions

Remark: An optimal pulse must balance finite duration and low spectral sidelobes—covered in lecture 2 (root-raised-cosine)

FT Example

$$\mathrm{rect}\left(\tfrac{t}{T}\right) \circ\!\!-\!\!\bullet\ T\,\mathrm{si}(\pi f T)$$

$$\mathrm{si}(x) = \frac{\sin(x)}{x}, \text{ zeros at } f = \pm\frac{k}{T}$$

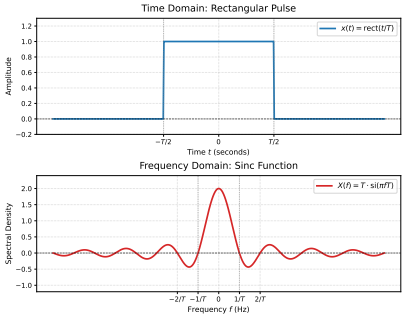


Figure: Rectangular Pulse and Si Function

Fourier Transformation III

Live demo: Frequency spectrum from the computer microphone

- scripts/section-1/live-spectrogram.py

Fourier Transformation IV

Essential Fourier Properties

- Time Shifting (e.g. Propagation Delay)
  - $x(t-\tau_0) \;\circ\!\!-\!\!\bullet\; X(f)\,\mathrm{e}^{-\mathrm{j}2\pi f\tau_0}$
  - Delay τ_0 introduces a linear phase rotation across frequency
- Frequency Shifting (e.g. Doppler Shift)
  - $x(t)\,\mathrm{e}^{\mathrm{j}2\pi f_\mathrm{D} t} \;\circ\!\!-\!\!\bullet\; X(f-f_\mathrm{D})$
  - Doppler shift f_D introduces linear phase rotation across time
- Convolution Theorem
  - $x(t) * h(t) \;\circ\!\!-\!\!\bullet\; X(f)\cdot H(f)$
  - Simplifies channel equalization from deconvolution to division
- Sampling / Periodicity Duality
  - Periodicity t/f $\Longleftrightarrow$ Sampling f/t

$$\underbrace{x_\mathrm{s}(t) = x(t)\cdot\sum_{n=-\infty}^{\infty}\delta(t-nT_\mathrm{s})}_{\text{Sampling in Time}} \quad \circ\!\!-\!\!\bullet \quad \underbrace{X_\mathrm{s}(f) = \frac{1}{T_\mathrm{s}}\sum_{k=-\infty}^{\infty} X\left(f-\frac{k}{T_\mathrm{s}}\right)}_{\text{Periodification in Frequency}}$$

Complex Baseband I

Why Complex Signals for Real Waves?

- Antennas emit and capture real-valued signals $s_{\mathrm{BP}}(t) \in \mathbb{R}$
- Communications signals usually narrowband ($B \ll f_{\mathrm{c}}$)
  - Carrier f_{c} fixed in the GHz regime (e.g., $3.5\,\mathrm{GHz}$)
  - Information encoded in amplitude $A(t)$ and phase $\phi(t)$
- Simulating carrier oscillations at GHz rates is computationally expensive

Equivalent Complex baseband (BB) Envelope

- Utilize complex numbers to combine amplitude and phase
- $s_{\mathrm{BB}}(t) = A(t)\,\mathrm{e}^{\mathrm{j}\phi(t)} \in \mathbb{C}$
- Describes signal independent of the carrier frequency

Bandpass-baseband transformation

Bandpass (BP) as real projection of BB signal

$$s_{\mathrm{BP}}(t) = \mathrm{Re}\Big\{ s_{\mathrm{BB}}(t)\,\mathrm{e}^{\mathrm{j}2\pi f_{\mathrm{c}} t} \Big\}$$

Advantages

- Spectrum centered around $0\,\mathrm{Hz}$
- Sample rate reduces from $2(f_{\mathrm{c}} + B/2)$ to B
- Clean mathematical framework for modulation and filtering

Complex Baseband II

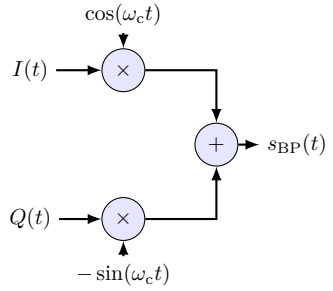
Cartesian Decomposition of BB

- Split complex envelope into real and imaginary parts:
- $s_{\mathrm{BB}}(t) = I(t) + \mathrm{j}Q(t)$
- $I(t) = \mathrm{Re}\{s_{\mathrm{BB}}(t)\} = A(t)\cos\phi(t)$ (**In-Phase**)
- $Q(t) = \mathrm{Im}\{s_{\mathrm{BB}}(t)\} = A(t)\sin\phi(t)$ (**Quadrature**)

Orthogonal Carrier Modulation

- Expanding $s_{\mathrm{BP}}(t)$ with Euler's identity
- $\mathrm{e}^{\mathrm{j}2\pi f_c t} = \cos\omega_c t + \mathrm{j}\sin\omega_c t$
- $s_{\mathrm{BP}}(t) = I(t)\cos(2\pi f_c t) - Q(t)\sin(2\pi f_c t)$
- Transmits two independent data streams over the same RF spectrum

in-phase and quadrature (I/Q) Modulator Architecture



Why do we have Digital Communication Systems?

Disadvantages of Analog Systems

- Requires two physical signal traces for I and Q
- Analog hardware impairments

Advantages of Digital Systems

- Complex numbers handled in memory/software
- Flexibility of the hardware platforms (Software-defined radio)
- Conditional access and encryption
- Error correction and information compression
- Storage of digital data

Discrete Fourier Transformation I

Discrete Counterpart of the FT

- The discrete Fourier transform (DFT) maps N discrete time samples to N discrete frequency bins
- DFT kernel from FT kernel
- $\mathrm{e}^{-\mathrm{j}2\pi ft} \quad \longrightarrow \quad \mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}kn} = W_N^{kn}$
- Discrete complex exponentials $\phi_k[n] = \mathrm{e}^{\mathrm{j}\frac{2\pi kn}{N}}$ serve as orthogonal basis

Intrinsic Periodicity of the DFT

- Discrete harmonic basis is N-periodic
- $W_N^{(k+N)n} = W_N^{kn} \implies X[k+N] = X[k]$
- Time sequence implicitly treated as periodic
- $\tilde{x}[n] = x[n \bmod N]$

N-Point DFT (Analysis)

Time sequence to discrete frequency

$$X[k] = \sum_{n=0}^{N-1} x[n]\, \mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}kn}$$

$$k \in \{0, 1, \ldots, N-1\}$$

N-Point inverse discrete Fourier transform (IDFT) (Synthesis)

Reconstruction of time samples

$$x[n] = \tfrac{1}{N} \sum_{k=0}^{N-1} X[k]\, \mathrm{e}^{\mathrm{j}\frac{2\pi}{N}kn}$$

$$n \in \{0, 1, \ldots, N-1\}$$

Orthogonality of Discrete Complex Exponentials

$$\left\langle \mathrm{e}^{\mathrm{j}\frac{2\pi kn}{N}}, \mathrm{e}^{\mathrm{j}\frac{2\pi mn}{N}} \right\rangle = \begin{cases} N, & k = m \\ 0, & k \neq m \end{cases}$$

Discrete Fourier Transformation II

- Time Shifting
  - Compare to FT (Slide 2.7)
  - $x\left[(n - n_0) \bmod N\right] = X[k]\, e^{-j\frac{2\pi}{N}kn_0}$
- Frequency Shifting
  - Compare to FT (Slide 2.7)
  - $X\left[(k - k_0) \bmod N\right] = x[n]\, e^{j\frac{2\pi}{N}k_0 n}$
- Convolution Discrepancy Compared to FT
  - Discrete LTI channels perform linear convolution
  - $y[n] = x[n] * h[n] = \sum_m h[m]x[n-m]$
  - Multiplying two DFTs yields circular convolution ($\circledast_N$)
  - $x[n] \circledast_N h[n] \circ\!\!-\!\!\bullet\, X[k] \cdot H[k]$

Circular Convolution $\circledast_N$

$$x[n] \circledast_N h[n] = \sum_{m=0}^{N-1} h[m]\, x[(n-m) \bmod N]$$

Discrete Fourier Transformation III

DFT is a linear transformation

- N-point DFT acts as matrix-vector product:

$$\mathbf{X} = \mathbf{F}_N \mathbf{x}$$

- $\mathbf{x} = [x[0], \ldots, x[N-1]]^T \in \mathbb{C}^N$
- $\mathbf{X} = [X[0], \ldots, X[N-1]]^T \in \mathbb{C}^N$
- $\mathbf{F}_N$ is a symmetric matrix with entries $[\mathbf{F}_N]_{k,n} = W_N^{k \cdot n}$

Orthogonality & IDFT

- Columns of $\mathbf{F}_N$ form an orthogonal basis:

$$\mathbf{F}_N^{\mathrm{H}} \mathbf{F}_N = N \mathbf{I}_N \implies \mathbf{F}_N^{-1} = \frac{1}{N} \mathbf{F}_N^{\mathrm{H}}$$

- IDFT matrix is conjugate transpose $\mathbf{F}_N^{\mathrm{H}}$ scaled by $1/N$

The DFT Matrix $\mathbf{F}_N$

$$\mathbf{F}_N = \begin{bmatrix} 1 & 1 & 1 & \ldots & 1 \\ 1 & W_N^1 & W_N^2 & \ldots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \ldots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \ldots & W_N^{(N-1)^2} \end{bmatrix}$$

- With $W_N = \mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}}$

Matrix-vector representation

- $\mathbf{X} = \mathbf{F}_N \mathbf{x}$
- $\mathbf{x} = \frac{1}{N} \mathbf{F}_N^{\mathrm{H}} \mathbf{X}$

Spatial Processing I

- Plane wave = oscillation across distance and time
- Wave number k_0 = spatial phase velocity per meter $(\mathrm{rad/m})$
- Wave vector k = directional vector pointing along direction of propagation θ
- Spatial sampling = antenna elements sample positions as analog-to-digital converter (ADC)s sample time
- Time–frequency duality maps to space–angle

Wave Basics

- Wave propagation in space and time
- $s(\mathbf{r}, t) = \mathrm{e}^{\mathrm{j}(\omega t - \mathbf{k}\cdot\mathbf{r})}$
- Spatial frequency vector
- $||\mathbf{k}||_2 = k_0 = \frac{2\pi}{\lambda}$

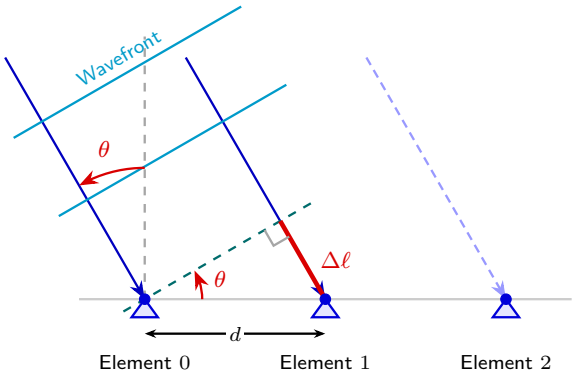
Time–Space Duality

Time Domain	**Spatial Domain**
Time t	Position x
Period T_s	Element Spacing d
Freq. ω	Wavenumber k_0
$f_\mathrm{s} \geq 2B$	$d \leq \lambda/2$

Spatial Processing II

Uniform linear array (ULA)

- Plane wave arriving from angle θ
- Wave travels extra Δl per element
- Elements experience angle-dependent phase shift ψ compared to reference element 0
- Array steering vector $\mathbf{a}(\theta)$
- $\mathbf{a}(\theta) = \begin{bmatrix} 1 & \mathrm{e}^{-\mathrm{j}\psi} & \dots & \mathrm{e}^{-\mathrm{j}(M-1)\psi} \end{bmatrix}^T$
- Spatial sampling without ambiguities if $d \leq \frac{\lambda}{2} \iff |\psi| \leq \pi$



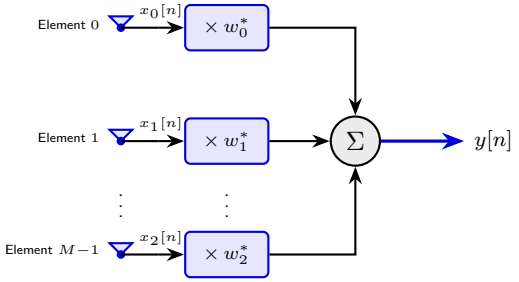
Path difference: $\Delta\ell = d \sin\theta$

Phase shift: $\psi = \frac{2\pi}{\lambda}\Delta\ell = \frac{2\pi d}{\lambda}\sin\theta$

Spatial Processing III

Bartlett Beamformer

- Weight vector $\mathbf{w}(\theta)$ "steers" antenna array
- Maximizes output power for direction θ by selecting weights as
- $\mathbf{w}_{\text{Bartlett}}(\theta) = \frac{1}{M}\mathbf{a}(\theta)$
- Undo phase shifts introduced by channel
- Scan output power of beamformer across θ
- $P_{\text{BF}}(\theta) = \mathbf{w}^H(\theta)\,\mathbf{R}_{xx}\,\mathbf{w}(\theta)$



Output Signal Model

- Snapshot vector: $\mathbf{x}[n] = \mathbf{a}(\theta_0)s[n] + \mathbf{n}[n]$
- Beamformer output: $y[n] = \mathbf{w}^H(\theta)\mathbf{x}[n]$
- Spatial covariance: $\mathbf{R}_{xx} = \mathbb{E}\{\mathbf{x}[n]\mathbf{x}^H[n]\}$

Spatial Processing IV

Live demo: Bartlett beamformer visualization

- scripts/section-1/bartlett.py
- scripts/section-1/bartlett-multi-doa.py

Standardization Preliminaries

Radio Spectrum

What is the Radio Spectrum?

- Part of electromagnetic spectrum from $3\,\mathrm{Hz}$ to $3\,\mathrm{THz}$
- Finite natural resource used for wireless communication
- Backbone of modern telecommunication and broadcasting

Spectrum Allocation

- Globally coordinated by International Telecommunication Union (ITU)
- Divided into frequency bands assigned to specific services (TV, radio, mobile, satellite, radar, amateur)

Frequency vs. Wavelength

- Connection between frequency f and wavelength λ
- $c = \lambda \cdot f$
- $c = 299\,792\,458\,\mathrm{m\,s^{-1}}$ (in vacuum)

Rules-of-Thumb

- $1\,\mathrm{GHz} \rightarrow 30\,\mathrm{cm}$
- $3\,\mathrm{GHz} \rightarrow 10\,\mathrm{cm}$
- $10\,\mathrm{GHz} \rightarrow 3\,\mathrm{cm}$

Radio Spectrum by ITU

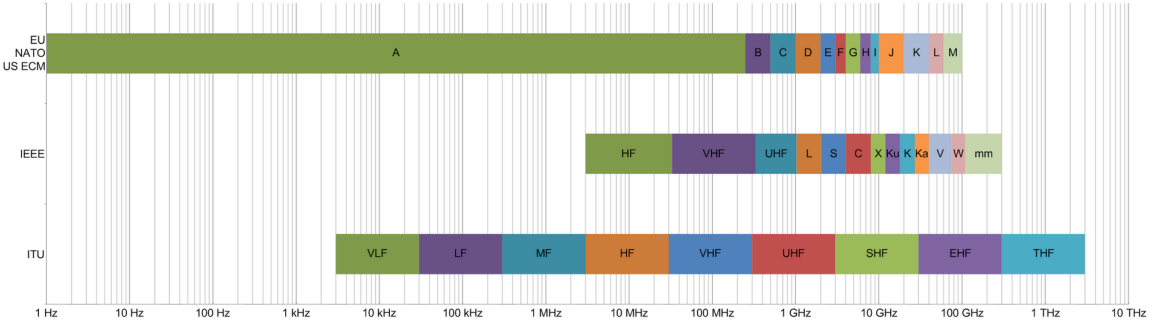


Figure: Radio Spectrum and ITU Frequency Bands [1]

Radio Spectrum Allocation USA

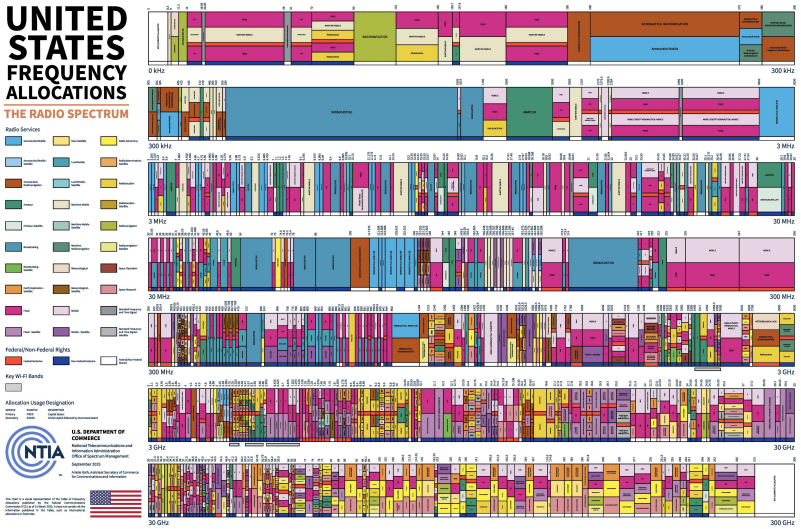


Figure: Radio Spectrum Allocation USA [2]

Allocation Example I

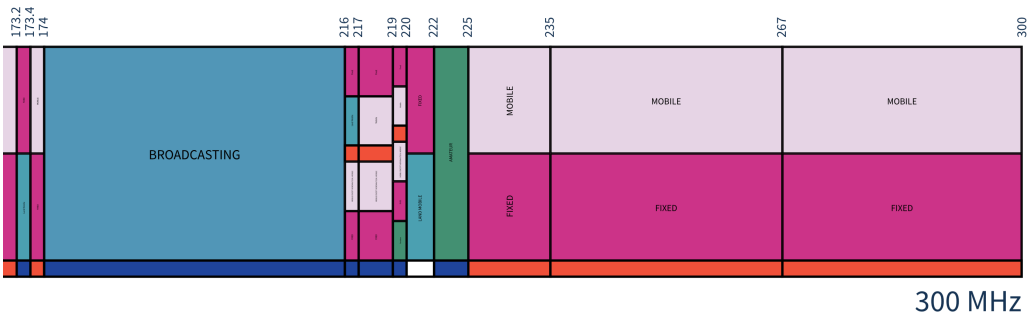


Figure: Radio Spectrum Allocation USA @300 MHz [2]

- Broadcasting starting @174 MHz: TV Channels
- Fixed/mobile starting @225 MHz: Military Band
  - Primary military (e.g., air-to-air/air-to-ground voice)
  - Aeronnautical radionavigation (instrument landing system (ILS) vertical guidance)

Allocation Example II

Frequency band $2.4\,\text{GHz}$ to $2.5\,\text{GHz}$

- Industrial, scientific and medical (ISM) band
  - High-power non-communication emitters (e.g., microwave ovens, industrial heating and drying)
  - Source of interference
- Unlicensed wireless systems
  - e.g., Wi-Fi, Bluetooth
  - Must tolerate ISM interference

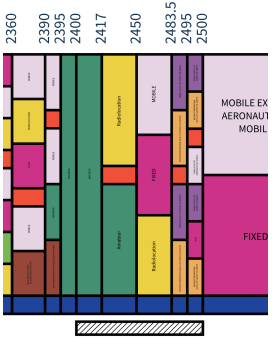


Figure: Radio Spectrum Allocation USA @$2.4\,\text{GHz}$ [2]

ITU

- United Nations Specialized Agency for Information and Communications Technology
- Founded in 1865 as the International Telegraph Union
- Renamed in 1947 to International Telecommunication Union
- Coordinates global use of the radio spectrum
- In total 194 member states
- Each member state has equal representation and voting rights e.g., on standards

ITU's three sectors

- ITU-R (Radio): Radio spectrum management, satellite coordination
- ITU-T (Telecommunication): Technical standards (Recommendations)
- ITU-D (Development): Digital development in member states

Standards

- Ensures fit for purpose, safe, and reliable products/services
- Promotes interoperability between equipment
- Harmonizes trade requirements across the globe
- Eliminates technical barriers
- Remark: Standards are adopted by mutual consensus; some become legally binding through regulatory reference (e.g., European Telecommunications Standards Institute (ETSI) standards cited in European Union (EU) law—USB-C in the EU common charger directive)

What is a standard?

A standard is an agreed-upon, repeatable technical specification or guideline designed to ensure that materials, products, processes, and services are safe, reliable, and mutually interoperable.

Key characteristics of radio standards

- Consensus-based: agreed by all parties
- Documented: published and accessible
- Voluntary: adoption is typically optional
- Continuously updated: revised as technology evolves

Standardization Bodies I

Standardization bodies develop, approve, and publish standards.

- International bodies
  - Coordinate global standards
  - e.g., International Organization for Standardization (ISO) (from Greek *isos*: equal)
  - e.g., ITU
- Regional bodies
  - Cover large geographic areas
  - e.g., ETSI
- National bodies
  - Represent each country
  - e.g., Deutsches Institut für Normung (DIN)
  - e.g., American National Standards Institute (ANSI)
- Industry consortia
  - Cover specific sectors
  - e.g., 3rd Generation Partnership Project (3GPP)
  - e.g., Institute of Electrical and Electronics Engineers (IEEE)

Role of industry consortia

- Standards of big alliances are directly adopted into official frameworks
- Consortium defines specs → adopted as regional standard → recognized globally
- e.g., Cellular standard (5G NR)

3GPP (TS 38.211)
↓
ETSI (TS 138 211)
↓
ITU-R (IMT-2020)

Standardization Bodies II

IEEE

- Global professional association
- Develops technical standards for electronics and computing
- e.g., IEEE 802.11 (Wi-Fi), 802.3 (Ethernet), 754 (floating-point)

ISO

- Network of national standards bodies
- Develops international standards across all industries
- e.g., ISO 9001 (quality), ISO 27001 (security), ISO/International Electrotechnical Commission (IEC) 14443 (RFID/NFC)

DIN

- German national standards body
- Representative for ISO/IEC
- DIN A4 paper, DIN 7258 (screws)

3GPP

- Collaboration of telecommunications standards organizations
- Develops specifications for mobile networks
- e.g., 3G, 4G, 5G, 6G

Standardization Process I

Types of standardization process

- Emergence as de facto standard (e.g., Ethernet)
- Written by standards organization (e.g., IEEE 802.11)
- Written by government or regulatory body (e.g., Federal Communications Commission (FCC) rules)
- Written by a corporation (e.g., Apple Lightning connector)

Standardize standards

The process of standardization can itself be standardized (e.g., ISO/IEC Directives Part 1/2).

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Figure: Excerpt from ISO/IEC Directives Part 1 [3]

Standardization Process II

Standards are versioned through releases, e.g., 3GPP

Release 8 (Long Term Evolution (LTE))
↓
Release 15 (5G)
↓
Release 21 (6G)

- Each release adds new features, fixes, and deprecations
- Industry coordinates on specific release versions
- Development can take years (especially for international bodies)
- Requires broad stakeholder participation
- Iterative process with multiple review rounds

Standardization Process (simplified)

1. Propose new topic or revision
2. Form working group of experts
3. Draft, review, and refine the specification
4. Achieve consensus and publish
5. Maintain and periodically review

Standardization Process III

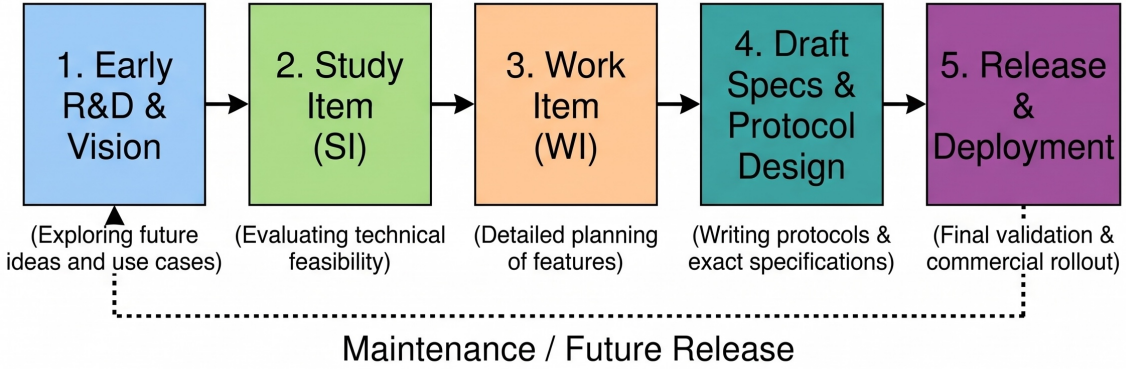


Figure: Standardization Process in 3GPP (simplified) [4]

Standardization Process IV

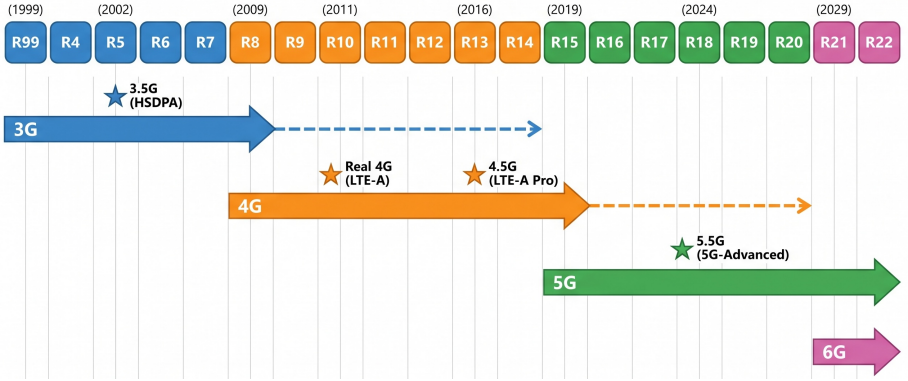


Figure: Releases in 3GPP over Time [4]

Patents I

Standards and patents intersect in radio systems, because standards often rely on patented innovations.

- Patent protects new technical solutions or products
- Grants patent holder monopoly for up to 20 years
- Prohibits making, using, or selling the invention
- However, inventor publicly discloses the invention

Requirements for patentability

- Novelty: Invention must be new
- Inventive step: Invention not obvious to skilled person
- Industrial application: Invention can be made or used

What is a patent?

A patent is an exclusive right granted for an invention.

Patent process (simplified)

1. File application with patent office
2. Formal examination (completeness)
3. Search for prior art
4. Substantive examination (novelty, inventiveness, applicability)
5. Grant or rejection
6. Maintain with annual renewal fees

Patents II

Standard-essential patent (SEP)

- Patents that claim technology necessary to implement a standard
- e.g., channel coding, multiple-input multiple-output (MIMO) schemes in 5G
- Any compliant device must use the patented technology
- Patent holder cannot exclude others from the standard market

Fair, reasonable, and non-discriminatory (FRAND) commitment

- SEP holders commit to license on FRAND terms
- Balances innovation incentives with standard adoption

FRAND principle

- Standardization increases patent value (must-use technology)
- FRAND limits royalty rates
- Tension between patent protection and open standards

3. Evolution of Cellular Standards

Introduction to Mobile Communication

What Makes Mobile Communication Hard?

Requirements

- Reach a user anywhere, without knowing where they are
- Keep a call alive while the user moves
- Serve many users at the same time
- Small, battery-powered terminals at low cost

Constraints

- Radio spectrum is a finite resource (slide 2.19)
- All users in an area share the same medium
- Terminals must work with the networks they encounter

Central Question

Why did early mobile communication develop towards **cellular** and **digital** systems?

Pattern of this Chapter

Each generation removes the bottleneck of the previous one.

Generations at a Glance

Generation	Era	Signal	Characteristic Technology
0G	from 1946	Analog	Single high site, few channels
1G	from 1979	Analog	Cellular reuse, FDMA
2G	from 1991	Digital	TDMA (GSM)
3G	around 2000	Digital	CDMA
4G	around 2010	Digital	OFDM (LTE)
5G	around 2020	Digital	OFDM (NR)
6G	around 2030 (expected)	Digital	In standardization

0G and the Capacity Problem

0G: Mobile Radio Telephony

United States [5]

- 1946: Public mobile telephone service in St. Louis
  - Three channels near 150 MHz
  - Calls handled by a mobile operator
- 1964: Improved Mobile Telephone System (IMTS)
  - Automatic channel selection
  - Customers dial themselves

Common Properties

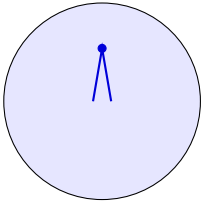
- Analog transmission
- Vehicle-mounted terminals
- One high site covers a whole city or region

Germany

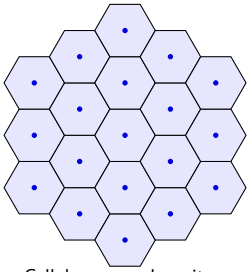
- 1958: A-Netz
  - Manually switched, peak of about $10\,800$ subscribers
- 1972: B-Netz
  - Direct dialing, peak of about $27\,000$ subscribers in 1986

0G: The Capacity Bottleneck

- One site occupies a channel in its entire coverage area
- Capacity = number of channels, regardless of area
- Only a few dozen channels available in many areas
- Demand exceeds supply: prospective customers end up on backlog lists



0G: one high site, channels used once



Cellular: many low sites, channels reused

Key Idea

Lower the transmit power and the same channel can be reused nearby.

The Cellular Concept

Cellular Concept I

Frequency Reuse

- Divide the service area into small cells
- Each cell uses a subset of the available channels
- A cluster of N cells uses all channels once
- Distant cells reuse the same channels
- Smaller N gives more channels per cell
- Concept formulated by D. H. Ring in 1947, first deployed decades later

Handover and Cell Splitting

- Handover: ongoing call moves to the next cell
- Cell splitting: smaller cells where demand grows, capacity scales with demand

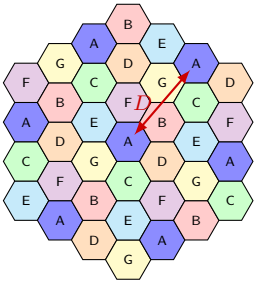


Figure: Reuse pattern for $N = 7$

Channels per Cell

- K total channels, cluster size N
- $k = K/N$ channels per cell

Cellular Concept II

Reuse Geometry

- Hexagonal cells with radius R (center to corner)
- Neighboring cell centers are $\sqrt{3}R$ apart
- Nearest co-channel cell: move i cells along one axis, turn by $60°$, move j cells
- Law of cosines with enclosed angle $120°$:

$$\begin{aligned} D^2 &= 3R^2 \left(i^2 + j^2 - 2ij\cos 120°\right) \\ &= 3R^2 \left(i^2 + ij + j^2\right) \end{aligned}$$

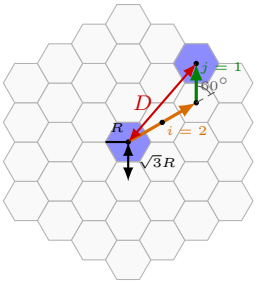


Figure: Nearest co-channel cell for $i = 2$, $j = 1$

Cluster Size

- Co-channel cells form a larger hexagonal grid with center spacing D
- Each large hexagon holds exactly one cluster
- Hexagon area scales with the square of the center spacing:

Co-Channel Reuse Ratio

- $N = i^2 + ij + j^2$
- $i, j \in \mathbb{N}_0$, $(i, j) \neq (0, 0)$
- $D/R = \sqrt{3N}$

Cellular Concept III

Co-Channel Interference

- Received power decays as $P \propto d^{-n}$ with path loss exponent n ($n = 2$ free space, $n \approx 3$–5 urban)
- First tier: six interferers at distance $\approx D$
- User at the cell edge, distance R to serving site
- Signal-to-interference ratio (SIR) depends only on $D/R = \sqrt{3N}$, independent of R and of the transmit power

Tradeoff: Small N gives more channels per cell ($k = K/N$) but a smaller D/R and thus a lower SIR.

Downlink SIR Estimate

$$\frac{S}{I} \approx \frac{R^{-n}}{6D^{-n}} = \frac{\left(\sqrt{3N}\right)^n}{6}$$

Examples for $n = 2$ and $n = 4$

		S/I in dB	
N	D/R	$n = 2$	$n = 4$
3	3.0	1.8	11.3
4	3.5	3.0	13.8
7	4.6	5.4	18.7
12	6.0	7.8	23.3

Cellular Concept IV

Live demo: SIR and capacity versus cluster size

- scripts/section-2/cellular-sir.py

1G: Analog Cellular Systems

Frequency Division Multiple Access

Frequency-division multiple access (FDMA)

- Each active user gets a dedicated frequency channel
- Continuous transmission for the whole call
- Simple receivers: filter out channel of interest
- Natural fit for analog frequency modulation (FM) voice

Consequences

- Channel width sets the capacity directly
- Guard bands and sharp filters required
- Base station needs one transceiver per channel

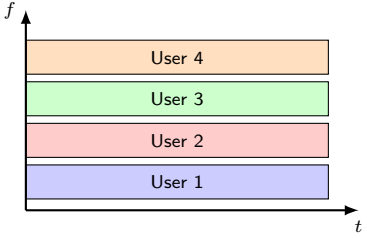


Figure: Time-frequency resources in FDMA

1G Systems

System	Region	Start
NTT	Japan	1979
NMT	Nordic countries	1981
AMPS	USA	1983
TACS	United Kingdom	1985
C-Netz	Germany	1985
Radiocom 2000	France	1985

Fragmentation in Europe

- National systems, mutually incompatible
- Terminals stop working at the border
- Small markets keep terminal prices high

- Exception: Nordic Mobile Telephone (NMT) was developed jointly and allowed roaming across the Nordic countries
- Germany: C-Netz grew to about $850\,000$ subscribers, far beyond A- and B-Netz
- Cellular solved capacity, but no system solved compatibility

Limits of 1G

- Analog voice degrades directly with interference
  - AMPS targets $18\,\mathrm{dB}$ over $90\,\%$ of the area
  - Requirement forces a large cluster size
- No confidentiality
  - Calls can be overheard with a scanner receiver
- Voice only
  - No native data or messaging service
- Limited options for further capacity
  - One transceiver per call
  - Speech cannot be compressed in analog form

Next Bottleneck

Cellular gives capacity by reuse, but the required SIR limits how tight the reuse can be.

2G: Digital Cellular Systems

Analog vs. Digital

	Analog (1G)	**Digital (2G)**
Voice	Frequency-modulated waveform	Compressed bit stream
Interference	Directly audible	Corrected up to a threshold
Tolerated SIR	about $18\,\mathrm{dB}$	about $9\,\mathrm{dB}$
Confidentiality	None	Encryption
Services	Voice	Voice, messaging, data
Base station	One transceiver per call	Several calls per transceiver
Terminals	Analog circuits	Digital signal processing

Remark: Digital processing became affordable only with progress in integrated circuits.

Time Division Multiple Access

Time-division multiple access (TDMA)

- Several users share one carrier in turns
- User transmits short bursts in an assigned time slot
- Requires digital signals: speech is buffered between bursts
- Requires time synchronization between users

Advantages over Pure FDMA

- One transceiver serves several calls
- Terminal does not transmit and receive simultaneously
- Idle slots allow measuring neighbor cells before handover

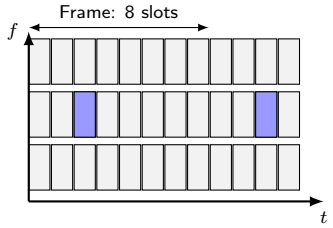


Figure: Combined FDMA/TDMA as used in Global System for Mobile Communications (GSM)

Remark: GSM is a hybrid: carriers separate in frequency, slots separate in time.

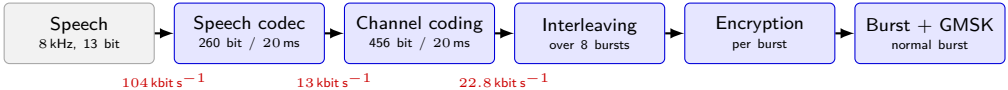
GSM Air Interface

Parameter	**Value (GSM 900)** [6]
Uplink band	$890\,\text{MHz}$ to $915\,\text{MHz}$
Downlink band	$935\,\text{MHz}$ to $960\,\text{MHz}$
Carrier spacing	$200\,\text{kHz}$
Carriers	124
Slots per carrier	8
Modulation	GMSK
Gross bit rate	$270.833\,\text{kbit}\,\text{s}^{-1}$

Bit Rate per User

- Carrier: $270.833\,\text{kbit}\,\text{s}^{-1}$
- Slot share: $\approx 33.9\,\text{kbit}\,\text{s}^{-1}$
- After burst overhead and control: $22.8\,\text{kbit}\,\text{s}^{-1}$
- Protects $13\,\text{kbit}\,\text{s}^{-1}$ speech with channel coding

From Speech to Burst



Unequal Error Protection [6]

Class	Bits	Protection
Ia	50	3 parity bits, convolutional code rate $1/2$
Ib	132	Convolutional code rate $1/2$
II	78	None

- $(50 + 3 + 132 + 4) \cdot 2 + 78 = 456$ bits per 20 ms
- 4 tail bits terminate the convolutional code
- Only bits important for speech quality are protected

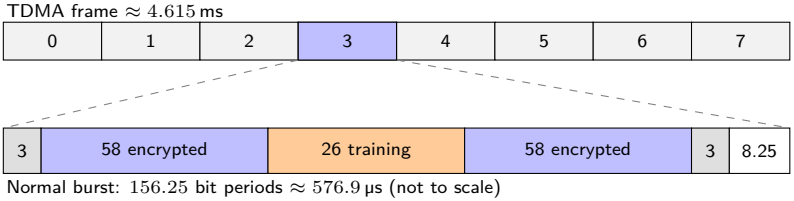
Interleaving

- 456 bits spread over 8 bursts, 57 bits each
- One lost burst only hits every 8th bit of a speech frame
- Scattered errors are corrected by the convolutional code

Key Insight

Coding and interleaving explain why GSM tolerates an SIR of about 9 dB.

GSM Frame and Burst Structure



- Training sequence: known bits in the middle of the burst, used to estimate the multipath channel and equalize
- Tail bits: define the start and end state of the equalizer
- Guard period: absorbs differences in arrival time between users of neighboring slots
- Timing advance: terminals far from the base station transmit earlier

Normal Burst

Bit Number	Length	Content
0–2	3	Tail bits
3–60	58	Encrypted bits (57 data, 1 stealing flag)
61–86	26	Training sequence bits
87–144	58	Encrypted bits (1 stealing flag, 57 data)
145–147	3	Tail bits
148–	8.25	Guard period

Normal burst for Gaussian minimum-shift keying (GMSK) [6]

GMSK

- Data is filtered by a Gaussian filter before frequency modulation
- Constant envelope modulation
- Amplifiers can run in saturation

- Specification = exact bit-level contract between independent manufacturers
- Rule of thumb: tables and figures first, prose second
- Task: find the slot duration and symbol rate in the specifications yourself

Why Different Bursts?

Normal Burst Assumes a Synchronized Terminal

- Carrier frequency known, oscillator corrected
- Slot and frame timing known
- Timing advance known, guard period only 8.25 bits

Terminal after Switch-On Knows None of This

- Where is a carrier, how far off is my oscillator?
  $\rightarrow$ Frequency correction burst (downlink)
- Where does the frame start, which frame, which base station?
  $\rightarrow$ Synchronization burst (downlink)
- How to reach the network without a timing advance?
  $\rightarrow$ Access burst (uplink)

Remark: Dummy bursts fill unused slots on the beacon carrier of each cell. Constant power allows neighbor cell measurements before handover.

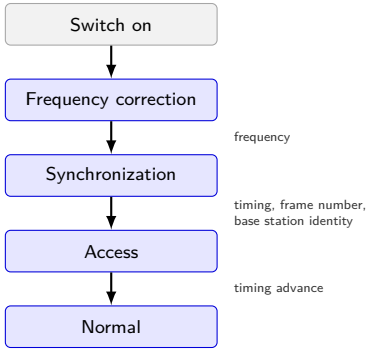
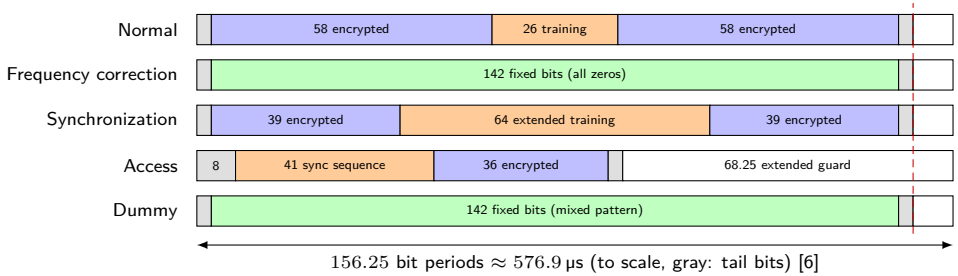


Figure: Bursts used by a terminal after switch-on

GSM Burst Types



- Frequency correction: all zeros make GMSK a pure tone $67.7\,\text{kHz}$ above the carrier, easy to find and to correct the oscillator
- Synchronization: extended training sequence for coarse timing, carries frame number and base station identity
- Access: timing advance still unknown, guard period extended to 68.25 bits
- Dummy: fixed pattern without user data, keeps the beacon carrier at constant power

Access Burst and Cell Size

Problem

- Base station expects bursts aligned to its slot grid
- Round-trip delay to a terminal at distance d: $\tau = 2d/c$
- Normal operation: timing advance compensates τ
- First access: timing advance unknown, burst arrives late by τ
- Normal guard period: 8.25 bits $\approx 30\,\mu s$, only $d \approx 4.6\,\text{km}$

Solution

- Short access burst of 88 bits, followed by a 68.25-bit guard period
- Late burst must still end within its own slot: $\tau \leq T_{\text{guard}}$
- Base station measures τ and assigns the timing advance

Remark: In GSM, timing sets an upper limit on the cell radius.

Maximum Cell Radius

Bit period [6]

$$T_{\text{b}} = \tfrac{48}{13}\,\mu s \approx 3.69\,\mu s$$

Guard period of access burst

$$T_{\text{guard}} = 68.25\,T_{\text{b}} \approx 252\,\mu s$$

Distance covered within guard period

$$d_{\text{max}} = \frac{c\,T_{\text{guard}}}{2} \approx 37.8\,\text{km}$$

Timing Advance Range

- 0 to 63 bit periods
- $d_{\text{max}} = \dfrac{c \cdot 63\,T_{\text{b}}}{2} \approx 35\,\text{km}$
- Effective limit $\approx 35\,\text{km}$: the

Timing Advance in Operation

Closed Control Loop [6]

1. Terminal derives uplink timing from downlink, uplink lags by 3 slots
2. Without timing advance (TA), bursts arrive late by $\tau = 2d/c$
3. Base station measures τ on the access burst via its sync sequence
4. Base station signals $\mathrm{TA} = \mathrm{round}(\tau/T_\mathrm{b}) \in \{0, \ldots, 63\}$ as 6 bit number
5. Terminal sends every burst $\mathrm{TA} \cdot T_\mathrm{b}$ earlier
6. Base station tracks arrival times via the training sequence and updates TA about every $480\,\mathrm{ms}$

Handover

- Distance to the new cell differs, TA is measured again with access bursts

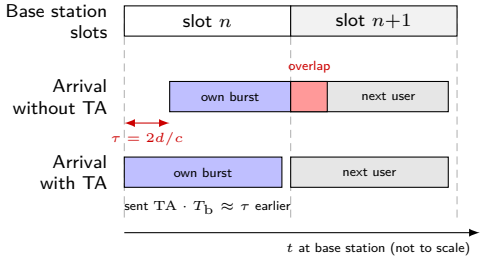


Figure: Burst arrival at the base station

Capacity Gain of Digital

Comparison for $n = 4$ (Cellular Concept III)

	AMPS	**GSM**
Required SIR	$\approx 18\,\text{dB}$	$\approx 9\,\text{dB}$
Cluster size N	7	3–4
Bandwidth per call B_{call}	$30\,\text{kHz}$	$200\,\text{kHz}/8 = 25\,\text{kHz}$
Calls per cell per MHz	≈ 4.8	≈ 10–13

- Calls per cell per MHz: $1/(B_{\text{call}} \cdot N)$
- Gain of about 2–3
- Mainly from smaller N ($k = K/N$), hardly from narrower channels

Interference Countermeasures

- Slow frequency hopping: new carrier every TDMA frame (≈ 217 hops/s), averages fading and interference [6]
- Power control: each link uses only the power it needs
- Discontinuous transmission: send only during speech activity

Key Insight

Digital gains capacity by tolerating more interference, which allows tighter reuse.

Data in 2G: GPRS and EDGE

Problem

- Circuit-switched data: $9.6\,\mathrm{kbit\,s^{-1}}$ in one slot
- Slot reserved for the whole connection, even when idle

General Packet Radio Service (GPRS)

- Packet switching: slots assigned only while data is queued
- One user may bundle several slots

Enhanced Data Rates for GSM Evolution (EDGE)

- Phase-shift keying (PSK) with 8 phases (8-PSK): 3 bits per symbol instead of 1 [6]
- Same symbol rate, same $200\,\mathrm{kHz}$ carrier, same slots
- Link adaptation up to $59.2\,\mathrm{kbit\,s^{-1}}$ per slot

	Modulation	kbit/s	
		Slot	8 slots
Circuit-sw.	GMSK	9.6	–
GPRS	GMSK	21.4	171.2
EDGE	8-PSK	59.2	473.6

Price of 8-PSK

No constant envelope, linear amplifiers required (compare GMSK on the Normal Burst slide)

Remark: $200\,\mathrm{kHz}$ carriers and TDMA slots remain the limit.

GSM Beyond the Air Interface

New Capabilities

- Subscriber identity module (SIM): separates subscriber from device
- Roaming across operators and borders
- Encryption of the radio link
- Short message service (SMS): first mass-market data service

Rollout

- First GSM call: 1 July 1991 in Finland
- Germany: D-Netz regular operation from 1992

Other 2G Systems

- D-AMPS (USA): TDMA on Advanced Mobile Phone Service (AMPS) channels
- PDC (Japan): TDMA
- IS-95 (USA): code-division multiple access (CDMA), covered with 3G

Key Insight

Roaming results from a common standard, not from digital technology.

ETSI and the GSM Standard

From CEPT to ETSI

Year	Milestone
1982	CEPT forms Groupe Spécial Mobile
1988	ETSI created (January)
1988	First GSM specifications ready
1989/90	GSM work moves to ETSI
1991	First GSM call
1998	3GPP formed

Why a New Body?

- European Conference of Postal and Telecommunications Administrations (CEPT) represented administrations
- ETSI also includes manufacturers and network operators
- Pan-European technology needs a pan-European process

Remark: GSM specifications are maintained in 3GPP today

ETSI and Patents: From SEP to FRAND

The Problem

- A standard may bestow a monopoly on one supplier
- GSM still contains many SEPs
- Motorola refused general licensing declarations and favored cross-licensing

Finding a Policy

- Early 1990s: licensing in advance as a condition of membership
- 1993: never in effect after criticism and a formal complaint
- November 1994: simplified intellectual property rights (IPR) policy approved

Result: ETSI IPR Policy

- Members declare essential patents
- Holders commit to FRAND licensing
- FRAND terms are not defined further

Scale

More than $23\,500$ patents declared essential to GSM

Summary: Why Cellular and Digital?

Shift	Bottleneck	Solution
Cellular (0G $\rightarrow$ 1G)	Channels usable only once per region	Frequency reuse in cells
Digital (1G $\rightarrow$ 2G)	High required SIR, no security	Coding, TDMA, encryption
Common standard	National, incompatible systems	GSM and ETSI
Packet data (2G $\rightarrow$ 2.5G)	Slot held for whole connection	GPRS, EDGE

Next: 3G

- Voice is solved
- Data demand is rising
- TDMA slots and $200\,\text{kHz}$ carriers limit the data rate

Is there another way to share the channel?

References I

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[6] ETSI, "Digital cellular telecommunications system (phase 2+) (GSM); GSM/EDGE physical layer on the radio path; general description (3GPP TS 45.001 version 18.0.0 release 18)," European Telecommunications Standards Institute (ETSI), Sophia Antipolis, France, Technical Specification (TS) 145 001, May 2024, Version 18.0.0.

Acronyms I

3GPP 3rd Generation Partnership Project. 34, 35, 37, 38, 39, 73

ADC analog-to-digital converter. 22
AMPS Advanced Mobile Phone Service. 56, 69, 71
ANSI American National Standards Institute. 34

BB baseband. 16, 17
BP bandpass. 16

CDMA code-division multiple access. 45, 71
CEPT European Conference of Postal and Telecommunications Administrations. 73
CIR channel impulse response. 11, 12
CTF channel transfer function. 12

DFT discrete Fourier transform. 19, 20, 21
DIN Deutsches Institut für Normung. 34, 35
DVB digital video broadcasting. 3

Acronyms II

EDGE Enhanced Data Rates for GSM Evolution. 70, 75

ETSI European Telecommunications Standards Institute. 33, 34, 73, 75

EU European Union. 33

FCC Federal Communications Commission. 36

FDMA frequency-division multiple access. 45, 55, 60

FM frequency modulation. 55

FRAND fair, reasonable, and non-discriminatory. 41, 74

FT Fourier transform. 12, 13, 19, 20

GMSK Gaussian minimum-shift keying. 61, 62, 64, 66, 70

GPRS General Packet Radio Service. 70, 75

GSM Global System for Mobile Communications. 45, 60, 62, 67, 69, 73, 74, 75

I/Q in-phase and quadrature. 17

ICAS integrated communications and sensing. 5

IDFT inverse discrete Fourier transform. 19, 21

Acronyms III

IEC International Electrotechnical Commission. 35, 36

IEEE Institute of Electrical and Electronics Engineers. 34, 35, 36

IFT inverse Fourier transform. 12

ILS instrument landing system. 30

IMTS Improved Mobile Telephone System. 47

IPR intellectual property rights. 74

ISM industrial, scientific and medical. 31

ISO International Organization for Standardization. 34, 35, 36

ITU International Telecommunication Union. 27, 28, 32, 34

LTE Long Term Evolution. 37, 45

LTI linear time-invariant. 11, 20

MIMO multiple-input multiple-output. 41

NMT Nordic Mobile Telephone. 56

NR new radio. 45

Acronyms IV

OFDM orthogonal frequency-division multiplexing. 45

PSK phase-shift keying. 70

SEP standard-essential patent. 41, 74
SIM subscriber identity module. 71
SIR signal-to-interference ratio. 52, 53, 57, 59, 62, 69, 75
SMS short message service. 71

TA timing advance. 68
TACS Total Access Communication System. 56
TDMA time-division multiple access. 45, 60, 69, 70, 71, 75

ULA uniform linear array. 23